

Linear Algebra of Quantum Mechanics

Methods and Models for Statistical Mechanics

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Hilbert Space — Definition

Definition

A **Hilbert space** $\mathcal{H}$ is a complex vector space with an inner product $\langle \cdot, \cdot \rangle$ satisfying:

- **Linearity:** $\langle \varphi, \alpha\psi_1 + \beta\psi_2 \rangle = \alpha\langle \varphi, \psi_1 \rangle + \beta\langle \varphi, \psi_2 \rangle$
- **Anti-linearity:** $\langle \alpha\varphi, \psi \rangle = \bar{\alpha}\langle \varphi, \psi \rangle$
- **Conjugate symmetry:** $\langle \varphi, \psi \rangle = \overline{\langle \psi, \varphi \rangle}$
- **Positive definiteness:** $\langle \varphi, \varphi \rangle \geq 0$, with $= 0 \Leftrightarrow \varphi = 0$
- **Completeness** w.r.t. the induced norm $\|\varphi\| = \sqrt{\langle \varphi, \varphi \rangle}$

Key Fact (finite dimensions)

In the finite-dimensional setting completeness is *automatic*: every finite-dimensional inner product space over $\mathbb{C}$ is a Hilbert space. Dimension N guarantees $\mathbb{C}^N \cong \mathcal{H}$ via any ONB.

Orthonormal Basis & Parseval Identity

Given $\dim \mathcal{H} = N$, an **orthonormal basis** (ONB) $\{e_n\}_{n=1}^N$ satisfies $\|e_n\| = 1$ and $\langle e_n, e_m \rangle = 0$ for $n \neq m$.

Fourier decomposition:
$$\varphi = \sum_{n=1}^N \alpha_n e_n, \quad \alpha_n = \langle e_n, \varphi \rangle$$

Parseval identity:
$$\langle \varphi, \psi \rangle = \sum_{n=1}^N \langle e_n, \varphi \rangle \langle e_n, \psi \rangle$$

Operator norm on $\mathcal{B}(\mathcal{H}, \mathcal{K})$

$\|A\| = \sup_{\|\varphi\|=1} \|A\varphi\|_{\mathcal{K}}$. Any A is uniquely determined by its matrix elements $a_{nm} = \langle f_m, A e_n \rangle$.

Adjoint & the C^* -Identity

Adjoint

For $A \in \mathcal{B}(\mathcal{H}, \mathcal{K})$, the adjoint $A^* \in \mathcal{B}(\mathcal{K}, \mathcal{H})$ is defined by

$$\langle \varphi, A\psi \rangle_{\mathcal{K}} = \langle A^*\varphi, \psi \rangle_{\mathcal{H}} \quad \forall \varphi \in \mathcal{K}, \psi \in \mathcal{H}.$$

Fundamental properties:

- Involution: $A^{**} = A$
- Norm preservation: $\|A\| = \|A^*\|$
- Anti-linearity: $(\alpha A)^* = \bar{\alpha} A^*$
- C^* -identity: $\|AA^*\| = \|A\|^2$

Significance

The C^* -identity $\|AA^*\| = \|A\|^2$ is the algebraic cornerstone distinguishing C^* -algebras from general Banach algebras; it encodes norm and algebraic structure simultaneously.

Distinguished Classes of Operators

Self-adjoint ($A = A^*$)

Real eigenvalues; model *observables*. Equivalent to $\langle \varphi, A\psi \rangle = \langle A\varphi, \psi \rangle$.

Unitary ($UU^* = I$)

Preserve inner products; model *symmetries*. In finite dims: $UU^* = I \Leftrightarrow U^*U = I$.

Normal ($AA^* = A^*A$)

$A = \frac{A+A^*}{2} + \frac{1}{2i}(A - A^*)$: sum of two self-adjoint operators.

Projection ($P^2 = P$)

Orthogonal if also $P = P^*$. Rank-one: $P = |\varphi\rangle\langle\psi|$ with $\langle\varphi|\psi\rangle = 1$; orthogonal $\Leftrightarrow \varphi = \psi$.

Spectral Theory — Key Concepts

Spectral theory underpins the measurement postulate: eigenvalues are outcomes; eigenprojections are events.

$\lambda \in \text{sp}(A) \Leftrightarrow (A - \lambda \mathbb{1})$ not invertible $\Leftrightarrow \lambda$ eigenvalue (finite dims).

$m(\lambda) = \dim E_\lambda$ (geometric mult.) $\mu(\lambda) = \dim G_\lambda$ (algebraic mult.)
 $m(\lambda) \leq \mu(\lambda)$; equality for self-adjoint ops.

Resolvent identity: $(z - A)^{-1} - (w - A)^{-1} = (w - z)(z - A)^{-1}(w - A)^{-1}$

Abstract Jordan Normal Form

$A = \sum_j (\lambda_j P_{\lambda_j} + D_{\lambda_j})$, coordinate-free: $P_{\lambda_j}^2 = P_{\lambda_j}$, $P_{\lambda_j} P_{\lambda_k} = 0$ ($j \neq k$),
 D_{λ_j} nilpotent.

Spectral Theorem — Self-Adjoint Operators

For $A = A^* \in \mathcal{B}(\mathcal{H})$ with $\text{sp}(A) = \{\lambda_1, \dots, \lambda_p\}$:

- All $\lambda_j \in \mathbb{R}$
- All eigenvalues *semi-simple*
($D_{\lambda_j} = 0$)
- Eigenprojections orthogonal
($P_{\lambda_j}^* = P_{\lambda_j}$)
- Eigenspaces mutually
orthogonal
- $\mathcal{H} = \bigoplus_j E_{\lambda_j}$

Spectral decomposition:

$$A = \sum_j \lambda_j P_{\lambda_j}$$

Functional calculus:

$$f(A) = \sum_j f(\lambda_j) P_{\lambda_j}$$

Properties of functional calculus

$$[f(A)]^* = \bar{f}(A), \quad (fg)(A) = f(A)g(A), \quad (f+g)(A) = f(A) + g(A).$$

Essential for defining e^A , $\log A$, A^p .

Completely Positive Maps & Quantum Channels

A **superoperator** $\Phi : \mathcal{B}(\mathcal{H}) \rightarrow \mathcal{B}(\mathcal{H})$ is *positive* if $X \geq 0 \Rightarrow \Phi(X) \geq 0$.
Positivity alone is **insufficient** for quantum-mechanical consistency.

Key example

The transpose map on $\mathcal{B}(\mathbb{C}^2)$ is positive but *not* 2-positive.

Kraus–Stinespring Theorem

Φ is **completely positive (CP)** $\iff$

$$\Phi(X) = \sum_{j=1}^m V_j X V_j^*, \quad V_j \in \mathcal{B}(\mathcal{H}).$$

Kraus operators unique up to unitary mixing; minimal $m = \text{rank } r(\Phi)$.
Equivalent (Stinespring): $\exists V : \mathcal{H} \rightarrow \mathcal{H} \otimes \mathbb{C}^r$ s.t. $\Phi(X) = V^*(X \otimes \mathbb{1})V$.

Quantum channel

A CP *unital* map $\Phi(\mathbb{1}) = \mathbb{1}$ (Heisenberg picture). Its adjoint Φ^* is the channel in the Schrödinger picture.

Perron–Frobenius Theory for CP Maps

The classical Perron–Frobenius theorem for non-negative matrices extends to positive maps on $\mathcal{B}(\mathcal{H})$.

Irreducible CP map

No non-trivial orthogonal projection P satisfies $\Phi(P) \leq \lambda P$. For such Φ :
 $\exists!$ strictly positive fixed point X of Φ^* with $\text{tr}(X) = 1$, and a spectral gap.

Primitive channels (period $p = 1$):

$$\|\Phi^{(n)}(X) - \text{tr}(XX) \cdot \mathbb{1}\| = \mathcal{O}(e^{-rn})$$

Significance

This quantum Perron–Frobenius theorem provides the foundation for the ergodic theory of quantum Markov chains, directly generalising the classical case.

Operator Monotone & Operator Convex Functions

Definition

$f : (a, b) \rightarrow \mathbb{R}$ is **operator monotone** if $A \geq B \Rightarrow f(A) \geq f(B)$ (operator ordering). *Strictly stronger* than scalar monotonicity.

Examples on $(0, \infty)$: $f(x) = -1/x$, $f(x) = x^p$ ($p \in [0, 1]$), $f(x) = \log(1+x)$.

Integral repr. ($0 < p < 1$): $x^p = \frac{\sin(\pi p)}{p} \int_0^\infty t^p \left[\frac{1}{t} - \frac{1}{t+x} \right] dt \Rightarrow A^p - B^p \geq 0$ whenever $A \geq B$.

Loewner's Theorem

$f : (0, \infty) \rightarrow \mathbb{R}$ is operator monotone $\Leftrightarrow f$ extends analytically to $\mathbb{C} \setminus (-\infty, 0]$ with the *Herglotz property*: $\text{Im}(f(z)) \geq 0$ whenever $\text{Im}(z) > 0$.

Key duality

f op.-convex on $(0, \infty)$, $f(0) = 0 \Leftrightarrow g(x) = f(x)/x$ op.-monotone.
 f op.-concave $\Leftrightarrow f$ op.-monotone.

Trace — Definition & Fundamental Properties

Definition

For any ONB $\{\varphi_n\}_{n=1}^N$ of $\mathcal{H}$:

$$\mathrm{tr}(A) = \sum_{n=1}^N \langle \varphi_n, A\varphi_n \rangle \quad (\text{independent of the ONB chosen}).$$

For self-adjoint $A = \sum_{\lambda} \lambda P_{\lambda}$: $\mathrm{tr}(A) = \sum_{\lambda} \lambda \dim(P_{\lambda})$.

Fundamental properties:

- **Linearity:**
 $\mathrm{tr}(\lambda A + \mu B) = \lambda \mathrm{tr} A + \mu \mathrm{tr} B$
- **Cyclicity:** $\mathrm{tr}(AB) = \mathrm{tr}(BA)$
- **Monotonicity:**
 $A \geq B \Rightarrow \mathrm{tr} A \geq \mathrm{tr} B$
- **Conjugate:** $\mathrm{tr}(A^*) = \overline{\mathrm{tr}(A)}$

Hilbert–Schmidt space

$\langle A, B \rangle = \mathrm{tr}(A^* B)$ makes $\mathcal{B}(\mathcal{H})$ a Hilbert space.

Superoperators $\Phi : \mathcal{B}(\mathcal{H}) \rightarrow \mathcal{B}(\mathcal{H})$ become operators on this inner product space, essential for quantum channel analysis.

Schatten p -Norms

The **absolute value** $|A| = \sqrt{A^*A} \geq 0$; its eigenvalues $\{\mu_j(A)\}$ are the *singular values*.

Schatten p -norm ($0 < p < \infty$): $\|A\|_p = (\operatorname{tr}|A|^p)^{1/p} = (\sum_j \mu_j(A)^p)^{1/p}$

$p = 1$: trace norm

$\|A\|_1 = \operatorname{tr}|A|$
(nuclear norm)

$p = 2$: HS norm

$\|A\|_2 = \sqrt{\operatorname{tr}(A^*A)}$
(inner product)

$p = \infty$: op. norm

$\|A\|_\infty = \max_j \mu_j(A)$

$p \mapsto \|A\|_p$ is real-analytic and decreasing in p , with $\lim_{p \rightarrow \infty} \|A\|_p = \|A\|_\infty$.

$|\operatorname{tr}(A)| \leq \|A\|_1$ (polar decomp. + Cauchy-Schwarz).

Hölder & Minkowski Inequalities for Schatten Norms

Hölder's Inequality

For $\frac{1}{p} + \frac{1}{q} = 1$, $p, q \in [1, \infty]$:

$$\|AB\|_1 \leq \|A\|_p \|B\|_q.$$

Proof via *Hadamard's three-line theorem*.

Minkowski's Inequality

For $p \in [1, \infty)$:

$$\|A + B\|_p \leq \|A\|_p + \|B\|_p.$$

Hence $\|\cdot\|_p$ is a norm on $\mathcal{B}(\mathcal{H})$ for $p \geq 1$.

Duality formula (key to Minkowski proof)

$\|T\|_p = \sup_{\|S\|_q \leq 1} |\operatorname{tr}(ST)|$. Minkowski follows:

$$\|A + B\|_p \leq \sup_{\|S\|_q \leq 1} (|\operatorname{tr}(SA)| + |\operatorname{tr}(SB)|) \leq \|A\|_p + \|B\|_p.$$

Trace Inequalities I — Peierls–Bogolyubov & Klein

Peierls–Bogolyubov Inequality

For self-adjoint A, B (quantum Jensen's inequality):

$$\log \frac{\operatorname{tr}(e^A e^B)}{\operatorname{tr}(e^B)} \geq \frac{\operatorname{tr}(A e^B)}{\operatorname{tr}(e^B)}.$$

Application: free-energy bounds in quantum statistical mechanics.

Klein's Inequality

For $A > 0, B > 0$ (operator analogue of $x \log(x/y) \geq x - y$):

$$\operatorname{tr}(A \log A - A \log B) \geq \operatorname{tr}(A - B), \quad \text{equality} \Leftrightarrow A = B,$$

where $\operatorname{tr}(A \log B) = \sum_j \log \lambda_j(B) \langle \psi_j | A | \psi_j \rangle$.

Significance

Klein's inequality is the foundation of **quantum relative entropy**:

$$S(\rho \| \sigma) = \operatorname{tr}(\rho \log \rho - \rho \log \sigma) \geq 0.$$

Trace Inequalities II — Golden–Thompson & ALT

Golden–Thompson Inequality

For self-adjoint A, B :

$$\mathrm{tr}(e^{A+B}) \leq \mathrm{tr}(e^A e^B), \quad \text{equality} \Leftrightarrow [A, B] = 0.$$

Proof via Lie–Trotter: $\lim_{n \rightarrow \infty} (e^{A/n} e^{B/n})^n = e^{A+B}$. Application: bounding partition functions in quantum stat. mech.

Araki–Lieb–Thirring (ALT) Inequality

For $A \geq 0, B \geq 0, p > 0, r \geq 1$:

$$\mathrm{tr}\left[(A^{1/2} B A^{1/2})^p\right] \leq \mathrm{tr}\left[(A^{r/2} B^r A^{r/2})^p\right].$$

Generalises Golden–Thompson; proves monotonicity of quantum relative entropies.

Summary of inequalities

Peierls–Bogolyubov · Klein (rel. entropy) · Golden–Thompson (partition fncs) ·
ALT (entropy monotonicity)

The Lattice of Quantum Events

Quantum events: self-adjoint projections $P \in \mathcal{B}(\mathcal{H})$, identified with closed subspaces via $P \leftrightarrow \text{Ran}(P)$.

Lattice $\mathcal{L}(\mathcal{H})$ operations:

- $P \leq Q$: $\text{Ran}(P) \subseteq \text{Ran}(Q)$
- $P \wedge Q$: proj. onto $\text{Ran}(P) \cap \text{Ran}(Q)$
- $P \vee Q$: proj. onto $\overline{\text{span}(\text{Ran}(P) \cup \text{Ran}(Q))}$
- $P^\perp$: proj. onto $\text{Ran}(P)^\perp$

Failure of Distributivity ($\dim \mathcal{H} \geq 2$)

$$P \wedge (Q \vee R) \neq (P \wedge Q) \vee (P \wedge R)$$

Counterexample in $\mathbb{C}^2$: three rank-one projections from non-collinear unit vectors give $P \wedge Q = 0$, $P \wedge R = 0$, $Q \vee R = \mathcal{H}$, so $P \wedge (Q \vee R) = P \neq 0 = (P \wedge Q) \vee (P \wedge R)$.

$\mathcal{L}(\mathcal{H})$ is an *orthocomplemented lattice*, but *not* a Boolean algebra: the mathematical origin of quantum incompatibility.

Density Matrices & Gleason's Theorem

Quantum Probability Measure (QPM)

$\mu : \mathcal{L}(\mathcal{H}) \rightarrow [0, 1]$ with $\mu(0) = 0$, $\mu(\mathbb{1}) = 1$, and additivity:
 $\mu(P_1 + \cdots + P_n) = \sum_j \mu(P_j)$ for mutually orthogonal $\{P_j\}$.

A **density matrix** $\omega \geq 0$ with $\text{tr}(\omega) = 1$ induces a QPM via $\mu(P) = \text{tr}(\omega P)$.

Gleason's Theorem (1957)

If $\dim \mathcal{H} \geq 3$ and μ is any QPM on $\mathcal{L}(\mathcal{H})$, then $\exists!$ density matrix ω such that $\mu(P) = \text{tr}(\omega P)$ for all P .

Consequences

- Density matrices are the *universal* quantum probability measures.
- No hidden-variable theory assigns definite values to all projections simultaneously (Kochen–Specker theorem).
- **Fails in dim 2:** QPMs on $\mathcal{L}(\mathbb{C}^2)$ not induced by any ω exist.

Bloch Sphere Parametrisation (dim = 2)

Density matrices on $\mathbb{C}^2$ correspond bijectively to vectors in the unit ball

$\mathbb{B} = \{\vec{a} \in \mathbb{R}^3 : |\vec{a}| \leq 1\}$ via:

$$\omega_{\vec{a}} = \frac{1}{2}(\mathbb{1} + \vec{a} \cdot \vec{\sigma}), \quad \vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z) \text{ (Pauli matrices).}$$

Pure states

Correspond to unit vectors $|\vec{a}| = 1$, forming the Bloch sphere $\mathbb{S}^2$.

$$\omega \text{ pure} \Leftrightarrow \omega^2 = \omega \Leftrightarrow \omega = |\psi\rangle\langle\psi|$$

Chaotic state

$\omega = \frac{1}{2}\mathbb{1}$ corresponds to $\vec{a} = \vec{0}$: maximum entropy,

$$S(\omega) = \log 2.$$

Gleason fails in dim 2

A QPM on $\mathcal{L}(\mathbb{C}^2)$ can be defined by a non-continuous function $\mathbb{S}^2 \rightarrow \{0, 1\}$ compatible with the lattice, which cannot equal $P \mapsto \text{tr}(\omega P)$ (always continuous).

Quantum Random Variables & Commutativity

A **quantum random variable (qrv)** is a self-adjoint $A \in \mathcal{B}(\mathcal{H})$. Given density matrix ω , the statistics of A are encoded in

$$P_A(\lambda_k) = \text{tr}(\omega P_{\lambda_k}), \quad A = \sum_k \lambda_k P_{\lambda_k}.$$

Commutativity and Classical Description

A and B commute $\Leftrightarrow$ they share a common ONB $\{\varphi_n\}$ with $A\varphi_n = \lambda_n\varphi_n$ and $B\varphi_n = \mu_n\varphi_n$. Both are then simultaneously described by the classical space $(\{1, \dots, N\}, P_{A/B})$.

Heisenberg Uncertainty Principle

For self-adjoint A, B and density matrix ω :

$$\sqrt{\text{Var}_\omega(A)} \cdot \sqrt{\text{Var}_\omega(B)} \geq \frac{1}{2} |\text{tr}(\omega [i(AB - BA)])|.$$

Non-commuting observables

If $[A, B] \neq 0$, no classical probability space simultaneously describes their joint statistics for all ω : *irreducible non-classicality*.

Quantum Marginals & Partial Trace

The tensor product $\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2$ replaces $\Omega_1 \times \Omega_2$. Summing over one subsystem becomes the **partial trace**.

Partial trace

$A_1 = \text{tr}_{\mathcal{H}_2}(A) \in \mathcal{B}(\mathcal{H}_1)$ is characterised by

$$\text{tr}_{\mathcal{H}}(A(X \otimes \mathbb{1})) = \text{tr}_{\mathcal{H}_1}(A_1 X) \quad \forall X \in \mathcal{B}(\mathcal{H}_1).$$

Explicitly: $\langle g_l | A_2 g_m \rangle = \sum_j \langle f_j \otimes g_l | A(f_j \otimes g_m) \rangle$.

Classical analogy fails: Entanglement

Bell state on $\mathbb{C}^2 \otimes \mathbb{C}^2$: $\psi = \frac{1}{\sqrt{2}}(e_1 \otimes e_2 + e_2 \otimes e_1)$. The pure QPM $\omega = |\psi\rangle\langle\psi|$ has:

$$\omega_1 = \omega_2 = \frac{1}{2}\mathbb{1} \quad (\text{chaotic state, maximal entropy}).$$

Pure composite state $\Rightarrow$ *maximally mixed* subsystems.

Entanglement & Von Neumann Entropy

Von Neumann Entropy

$$S(\omega) = -\text{tr}(\omega \log \omega) = -\sum_j \lambda_j(\omega) \log \lambda_j(\omega).$$

Special values:

- Pure state $\omega = |\psi\rangle\langle\psi|$:
 $S(\omega) = 0$
- Chaotic state $\omega = \frac{1}{N} \mathbb{1}$:
 $S(\omega) = \log N$ (max)
- Bell state marginal:
 $S(\omega_1) = S(\omega_2) = \log 2$

Entanglement entropy

For pure $\psi \in \mathcal{H}_1 \otimes \mathcal{H}_2$:

$$E_{\text{ent}}(\psi) = S(\omega_1) = S(\omega_2).$$

Bell state: $E_{\text{ent}} = \log 2$ (max. for qubit).

Interpretation

Entanglement is a purely quantum-informational resource with no classical analogue: complete knowledge of the composite system is compatible with maximal uncertainty about each subsystem.

Classical vs. Quantum Stochastic Processes

Classical Stochastic Process

Family $(\Omega^N, P_N)_{N \geq 1}$ with compatibility:

$$P_N(\omega) = \sum_{\omega'} P_{N+M}(\omega, \omega').$$

Specific entropy:

$$s = \lim_N \frac{1}{N} S(P_N) = \inf_N \frac{S(P_N)}{N}.$$

Quantum Stochastic Process

Family $(\mathcal{H}^N, \omega_N)_{N \geq 1}$ with compatibility:

$$\omega_N = \text{tr}_{\mathcal{H}_M}(\omega_{N+M}).$$

i.i.d.: $\omega_N = \rho^{\otimes N}$; specific entropy $= S(\rho)$.

Both satisfy **subadditivity**: $S(\omega_{N+M}) \leq S(\omega_N) + S(\omega_M)$; the specific entropy $s = \lim_N \frac{1}{N} S(\omega_N)$ exists and equals the infimum.

Connection to quantum measurement

When $\mathcal{U} = C(\Omega)$ and $\mathcal{R} = \mathcal{B}(\mathcal{H})$, the QMC reduces to a classically-indexed process via quantum instruments $\{E_\omega\}_{\omega \in \Omega}$: statistics of repeated measurements.

Quantum Markov Chains & C^* -Algebraic Framework

QMCs are constructed over C^* -algebras $\mathcal{U}$ (observable) and $\mathcal{R}$ (ancilla):

Transition kernel $E : \mathcal{U} \otimes \mathcal{R} \rightarrow \mathcal{R}$ (CP unital) and **stationary state** ρ on $\mathcal{R}$ with $\rho(E(\mathbb{1}_{\mathcal{U}} \otimes B)) = \rho(B)$.

QSP: $\omega_N(A_1 \otimes \cdots \otimes A_N) = \rho(E_{A_1} \circ \cdots \circ E_{A_N}(\mathbb{1}_{\mathcal{R}}))$.

Gelfand–Naimark classification

Every finite-dim. C^* -algebra $\cong \bigoplus_k \mathcal{B}(\mathcal{H}_k)$. Commutative C^* -algebra $\cong C(\Omega) \Leftrightarrow$ classical probability space.

Ergodic theorem for primitive channels

For a primitive (aperiodic) irreducible CP channel ϕ :

$$\phi^n(A) \xrightarrow{n \rightarrow \infty} \text{tr}(A) \cdot X \quad \text{exponentially fast,}$$

the quantum analogue of the classical Markov chain ergodic theorem.

Concluding Remarks

Three themes woven throughout the material:

Algebraic and Analytic Interplay

Interplay of C^* -algebras, tensor products, CP maps with trace inequalities, operator monotone functions, and spectral theory.

Classical and Quantum Probability

Quantum probability reduces to classical for commuting observables; the C^* -algebraic framework precisely demarcates the boundary.

Mathematics and Physics

Quantum measurement, entropy, channels, and statistical mechanics motivate every mathematical choice throughout.

Prerequisites for advanced topics

Quantum hypothesis testing · Quantum Fisher information · Entropy inequalities · Open quantum systems · Quantum error correction.

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Proof & Exercises

(to be presented separately)